Matteo De Francesco

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Research Profile

I am a final-year PhD candidate in Theoretical Computer Science at the University of Geneva, working at the interface of graph theory and quantum computing. Over the course of my PhD I have moved from the probabilistic analysis of temporal graph algorithms, to quantum algorithm design on neutral-atom hardware, and I am now focusing on graph states and measurement-based quantum computation (MBQC), while continuing to investigate dynamic graph embeddings for neutral-atom machines. I enjoy quickly acquiring new theoretical frameworks and connecting them to my existing background in graph theory and probabilistic methods.

Research Interests

Graph States & Measurement-Based QC	Quantum Error Correction
Quantum Information	Quantum Algorithms
Distributed Quantum Computing	Quantum Optimization (QAOA)
Graph Theory	Computational Complexity

Education

PhD student in Computer Science — Jan. 2024 - ongoing
University of Geneva, Switzerland

Master's Degree (MSc) in Computer Science — Sep. 2020 - Feb. 2023
University of Pisa, Italy — 110/110 with honors

Bachelor's Degree (BSc) in Computer Science — Sep. 2017 - Jul. 2020
University of Pisa, Italy — 104/110

Research/Work Experience

PhD Student — Jan. 2024 - ongoing
University of Geneva

* Researcher in the ALGO team (supervised by Arnaud Casteigts) on quantum algorithms and graph theory.
* I worked on algorithms for motion planning problems and probabilistic process in random graphs, using martingale methods, Azuma's inequality, and Poisson process couplings.
* Investigating the generation of dynamical graph instances for neutral atom machines.
* Currently focusing on graph states and measurement-based quantum computation, building on my graph-theoretic background.
* Teaching assistant for the courses of "Formal Languages," "Computability and Complexity", "Algorithms" (Bachelor's), and "Semantics Modelling and Verification," "Graph Algorithms" (Master's).

Quantum Research Intern — Mar. 2023 - Oct. 2023
PASQAL

* Quantum research internship at PASQAL, a neutral-atom quantum computing company founded by 2022 Nobel Laureate in Physics Alain Aspect.
* Designed novel qubit register strategies to embed QUBO problems on neutral-atom QPUs, improving hardware utilization for combinatorial optimization.
* Built a hybrid classical-quantum column generation pipeline for large-scale optimization problems, combining QAOA with classical OR solvers.
* Gained hands-on experience with quantum algorithm design, Hamiltonian encodings, and quantum optimization on real neutral-atom hardware.

Research Intern — Sep. 2022 - Feb. 2023
EdF Lab R&D Paris-Saclay

* Designed an HPC-based algorithmic framework using Bundle methods for a class of large-scale optimization problems, with a theoretical extension supporting multi-capability cooperating solvers.
* Implemented and benchmarked the framework in C++ on the Unit Commitment problem, demonstrating measurable timing improvements.

Research Fellow — Apr. 2022 - Aug. 2022
University of Pisa

- Fully-funded research scholarship from the Italian Ministry of Justice ("Giustizia Agile" EU project); modernized ICT infrastructure for the law courts of Leghorn, Lucca, and Pisa.
- Built an interactive web application (Flask, ElasticSearch) for storing, searching, and visualizing court records, and prototyped NLP techniques for document analysis.

Teaching Assistant — Multiple periods
University of Pisa

- Selected three times as teaching assistant for "Fundamentals of Computer Science" and "Mobile Application Development" (Bachelor's level).
- Supported students on basic set operations, discrete mathematics, and graph theory; built an Android teaching app (Kotlin).

Publications & Research Projects

Optimal Trajectories in Discrete Space with Acceleration Constraints — 2026

Publication: A. Casteigts, M. De Francesco, P. Leone. arXiv:2602.21964. Study of optimal trajectories in discrete space under acceleration constraints, including efficient algorithms for trajectory optimization.

Randomized Algorithms for Temporal Spanners — Ongoing

Developing a randomized bottom-up algorithm for recovering a $2n-3$ temporal spanner, with theoretical analysis of its correctness and performance.

Graph Embeddings for Neutral-Atom Quantum Computing — Ongoing

Exploring how graph instances can be embedded and dynamically transformed on neutral-atom quantum machines, together with algorithmic approaches based on QAOA and related quantum optimization techniques.

Fermionic Advantange for LOCAL problems — Ongoing

Investigating how fermionic particles could lead to an advantage over quantum information in the LOCAL model of distributed quantum computing.

Graph States for Measurement-Based Quantum Computation — Ongoing

Studying the structure of graph states and their role as a computational resource in measurement-based quantum computation, with an eye toward connections to quantum error correction and temporal graphs.

Awards & Scholarships

2022	Fully-funded research scholarship, Italian Ministry of Justice ("Giustizia Agile")
2023	MSc in Computer Science, University of Pisa – 110/110 *with honors*

Technical Skills

Quantum	QAOA, Neutral-atom QPUs, Quantum optimization, QuTiP
Theory	Computational complexity, Algorithm design, Probabilistic analysis, Operations Research
Languages	Python, C++, Julia, MATLAB, C, Java, Bash
Frameworks	PyTorch, TensorFlow, CPLEX, Anaconda, Git

Languages

Italian	Mother tongue
English	Advanced
French	Advanced